

Uniform real-rootedness of the $(132, 213)$ -avoiding permutation-polytope family

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Abstract

We prove that the Ehrhart h^* -polynomial of the convex hull of the $(132, 213)$ -avoiding permutations in S_d has only negative real zeros for every $d \geq 3$. A uniform analytic argument proves positivity of the repaired Eulerian tails for every $d \geq 1001$. Combining this with de Castro's finite Theorem 1.3 for $d \leq 1000$ resolves Conjecture 10.1 of arXiv:2609.06096v1. The new argument uses centered Bernoulli-sum comparison, finite centered moments of an infinite correction series, and degree smoothing for its large components. Exact rational certificates establish scalar envelopes on 33 closed intervals and the finite bases of infinite inductions. The original finite dataset is an explicitly cited prior input, not an independently reconstructed computation. A short corollary applies Zhang's known Veronese preservation theorem to all positive integral dilates of this particular family.

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1 Statement, dependencies, and notation

Let P_d be the convex hull of the permutations in S_d avoiding both 132 and 213, in its induced affine lattice, and write $H_d(t) = h^*(P_d; t)$. The subscripted polynomials $P_n(z)$ below denote primitive auxiliary products, not the polytope P_d .

Theorem 1.1. *For every integer $d \geq 3$, all zeros of H_d are real and strictly negative. For $d = 1, 2$, $H_d = 1$.*

The new analytic result in this paper covers *every* $d \geq 1001$. The remaining dimensions use the prior finite Theorem 1.3 of de Castro [1]. We do not claim a new general preservation theorem for other pattern classes, poset permutahedra, or arbitrary lattice polytopes.

We use the following precise prior inputs from the fixed version [1]. Its refined Eulerian coordinates $a_{d,j}$ satisfy the exact transform in its equation (25). Its Corollary 6.2 says that nonnegative endpoint coordinates together with

$$S_{d,r}(c) = \sum_{j=r}^{d-2} c^{j-r} a_{d,j} \geq 0 \quad (1 \leq r \leq d-2), \quad c = 2 - \sqrt{3}, \quad (1)$$

imply real-rootedness for $d \geq 4$. The endpoint coordinates are nonnegative, with $a_{d,0} = 1$ and positive terminal coordinate for $d \geq 3$. Its Theorem 1.3 supplies $d \leq 1000$. The criterion is sometimes called Proposition 6.2 elsewhere in that source; we refer to its displayed Corollary 6.2.

The primitive and divisor polynomials, and the signed correction series, are

$$P_n(z) = \frac{1}{n^2(n-1)!} \prod_{q=1}^{n-1} (2n - q + qz), \quad (2)$$

$$J_n(z) = \sum_{h|n} \frac{\varphi(h)}{h^2} (z-1)^{n-n/h} P_{n/h}(z), \quad (3)$$

$$J(u, z) = \sum_{n \geq 1} J_n(z) u^n, \quad \mathcal{H}(u, z) = e^{(z-1)J(u,z)} = \sum_{s \geq 0} \mathcal{H}_s(z) u^s. \quad (4)$$

Their exact marked identity is

$$\mathcal{R}_d(z) := \sum_{m=0}^{d-1} a_{d,d-1-m} z^m = \sum_{s=0}^{d-1} \frac{d-s}{d} J_{d-s}(z) \mathcal{H}_s(z). \quad (5)$$

These identities are equations (34)–(37) of [1]. For $x > 0$ let

$$f_x(t) = \frac{xt}{2 + (x-1)t}, \quad p = \frac{x}{1+x}, \quad \nu = \int_0^1 f_x(t) dt, \quad (6)$$

$$X_{n,x} = \sum_{q=1}^{n-1} \text{Bernoulli}(f_x(q/n)), \quad \mu_n = \mathbb{E}X_{n,x}, \quad V_n = \text{Var } X_{n,x}. \quad (7)$$

All Bernoulli summands are independent. In particular,

$$[z^j]P_n(z)x^j = P_n(x)\mathbb{P}(X_{n,x} = j), \quad \frac{\mu_n}{1+x} \leq V_n \leq \mu_n. \quad (8)$$

The mean $\mu_d(x)$ increases continuously and strictly from 0 to $d-1$; hence $1 \leq m \leq d-2$ determines a unique saddle $\mu_d(x) = m$.

Set

$$\lambda = \exp\left(1 + \int_0^1 \log(2 + (x-1)t) dt\right), \quad R = \lambda^{-1}, \quad q = (1+x)R, \quad E = \mathcal{H}(R, x). \quad (9)$$

We also use the following product and norm bounds from Lemma 9.1 of [1], uniformly for $0 \leq x \leq 15$:

$$P_n(x)R^n = \frac{e^{\delta_n}}{\sqrt{4\pi(1+x)}n^{5/2}}, \quad -1/n < \delta_n < 0 \quad (n \geq 2), \quad (10)$$

$$\lambda \geq 4, \quad q \leq 4/5, \quad \|J_n - P_n\|_x R^n \leq (4/5)^{n/2} \quad (n \geq 1). \quad (11)$$

Here $\|F\|_x = \sum_j |[z^j]F|x^j$. Darroch's theorem [2], in the form reviewed in [3, p. 284], identifies an integer mean of a finite Bernoulli sum with its largest atom. No prior eventual-dimension conclusion is needed below.

2 The forward repaired tails

We sharpen only the mass constant in the forward estimate of [1, Lemmas 8.1–8.3 and Proposition 8.4]. Write

$$L_n(x) = \frac{1}{n^2(n-1)!} \prod_{q=1}^{n-1} (xn + q), \quad \mathcal{E}(x) = \frac{(1+x)^{1+x}}{x^x}.$$

For $5/54 \leq x \leq 2$, the Robbins and Riemann bounds give

$$L_n(x)\mathcal{E}(x)^{-n} \leq \frac{n^{-5/2}}{x\sqrt{2\pi}} < \frac{9}{2}n^{-5/2} \quad (n \geq 2).$$

At $n = 1$ the normalized term is $q_F = \mathcal{E}(x)^{-1} < 49/64$. Since $\sum_{n \geq 2} n^{-5/2} < 3/8$ and $\varphi(h)/h^2 \leq 1/2$ for $h \geq 2$, the normalized connected mass satisfies

$$A < \left(\frac{49}{64} + \frac{9}{2} \frac{3}{8}\right) \left(1 + \frac{q_F}{2(1-q_F)}\right) < \frac{157}{64} \frac{79}{30} = \frac{12403}{1920} < \frac{13}{2}.$$

The connected series is $K(u, x) = \sum_n k_n(x)u^n$, with $k_n(x) = \sum_{h|n} \varphi(h)h^{-2}L_{n/h}(x)$, and $\mathcal{F}_d(x) = [u^d]e^{K(u,x)}$. Its coefficients are nonnegative. The prior connected coefficient bound $5000n^{-5/2}$ remains valid. For nonnegative logarithmic coefficients bounded by $Cn^{-5/2}$ and mass A , a k -part

composition has a part at least n/k ; freely summing the others yields $Ck^{7/2}A^{k-1}n^{-5/2}$. Summing its exponential weights, using $k^{7/2} \leq k^4$, gives

$$[u^n] \exp\left(\sum b_j u^j\right) \leq Ce^A(A^3 + 6A^2 + 7A + 1)n^{-5/2}.$$

Consequently the prior $\exp(140)$ envelope can be replaced by $\exp(30)$:

$$5000e^A(A^3 + 6A^2 + 7A + 1) < 5000 \cdot 687e^{13/2} < 2^{22}e^{13/2} < e^{30}.$$

Here is the explicit tail comparison to which the improved envelope applies. Let $y = \sqrt{3} - 1$, $q_C = 5/113$, and $x_C = 5/54$. The refined transform gives

$$\mathcal{A}_d(z) = \sum_j a_{d,j} z^j = (1 - z)^{d-1} \mathcal{F}_d(2z/(1 - z)).$$

Its alternative polynomial expansion in the nonnegative coefficients of $\mathcal{F}_d$ bounds it on $|z| = q_C$ by $(1 + q_C)^{d-1} \mathcal{F}_d(x_C)$. Put $\mathcal{M} = c^{-r} y^{d-1} \mathcal{F}_d(y) > 0$. Subtracting the initial coordinates and the terminal coordinate gives

$$S_{d,r}(c) = \mathcal{M} - c^{-r} \sum_{j < r} c^j a_{d,j} - c^{d-1-r} a_{d,d-1}.$$

Cauchy's coefficient bound and the geometric sum show that the relative absolute initial-coordinate loss is at most

$$\mathcal{R} = \frac{q_C}{c - q_C} (c/q_C)^r ((1 + q_C)/y)^{d-1} \frac{\mathcal{F}_d(x_C)}{\mathcal{F}_d(y)}.$$

The terminal loss is at most $\mathcal{U} = (c/y)^{d-1} \mathcal{F}_d(2)/\mathcal{F}_d(y)$, since $0 < a_{d,d-1} \leq \mathcal{F}_d(2)$. The prior primitive lower bound is $\mathcal{F}_d(y) \geq d^{-5/2} \mathcal{E}(y)^d/36$. The exact inequalities of [1, Lemma 8.3] are

$$(c/q_C)^{1/4} \frac{1 + q_C}{y} \frac{\mathcal{E}(x_C)}{\mathcal{E}(y)} < 19/20, \quad \frac{c}{y} \frac{\mathcal{E}(2)}{\mathcal{E}(y)} < 77/100.$$

For $r \leq d/4$, insert the improved upper envelope and use $q_C(c - q_C)^{-1}y/(1 + q_C) < 1$ and $y/c < 3$. The two relative losses are therefore bounded by $36e^{30}(19/20)^d$ and $108e^{30}(77/100)^d$. All intermediate bounds in that argument hold already for $d \geq 2$; its old dimension threshold arose only from these final errors. For $d \geq 1001$, $e < 4$, $(19/20)^{14} < 1/2$, and $(77/100)^3 < 1/2$ give respective bounds 2^{-5} and 2^{-266} . Their sum is below one.

Proposition 2.1. *For $d \geq 1001$ and $1 \leq r \leq \lfloor d/4 \rfloor$, $S_{d,r}(2 - \sqrt{3}) > 0$.*

3 Centered local comparison

3.1 Fourier profiles and a modal lower bound

For a finite Bernoulli parameter vector $\mathbf{r}$, of mean μ and variance V , define its centered characteristic function and real profile

$$\psi(t) = \mathbb{E}e^{it(X-\mu)}, \quad F(y) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \psi(t) e^{-iyt} dt.$$

The profile is real by conjugate symmetry. It equals a point probability only when $y + \mu$ is an integer; no probability interpretation at other arguments is needed.

For one parameter put $g_r(t) = e^{-irt}(1 - r + re^{it})$. Taylor's integral remainder gives

$$|\partial_r g_r(t)| \leq |e^{it} - 1 - it| + r|t||e^{it} - 1| \leq \frac{3}{2}t^2.$$

If the other summands in each replacement have variance at least $W \geq 0$, their characteristic product is bounded by $e^{-2W \sin^2(t/2)}$. Pad the shorter vector by zero parameters and let $D = \sum_i |p_i - q_i|$. Telescoping proves

$$|F_p(y) - F_q(y)| \leq \frac{3}{2}DC_2(W), \quad C_2(W) = \frac{1}{2\pi} \int_{-\pi}^{\pi} t^2 e^{-2W \sin^2(t/2)} dt. \quad (12)$$

Apply replacements with nonnegative variance increments first, then negative ones. Every intermediate total variance is at least the smaller endpoint variance. Removing one coordinate costs at most $1/4$, so one may take $W = \max(0, \min(V_p, V_q) - 1/4)$. For $W > 0$, using $\sin(t/2) \geq |t|/\pi$ yields

$$C_2(W) \leq \min \left(\frac{\pi^2}{3}, \frac{\pi^{5/2}}{4 \cdot 2^{3/2}} W^{-3/2} \right). \quad (13)$$

Linear sine terms in $\text{Im } g_r$ cancel, giving $|\text{Im } g_r(t)| \leq r(1 - r)|t|^3/6$. Telescoping a product against its conjugate gives $|\text{Im } \psi(t)| \leq (V/6)|t|^3 e^{-2(V-1/4)\sin^2(t/2)}$ when $V > 1/4$. Differentiating the finite integral therefore proves

$$|F(\Delta) - F(0)| \leq A_1(V)|\Delta| + \frac{1}{2}C_2(V)\Delta^2, \quad (14)$$

$$A_1(V) \leq V\pi^4/30 \quad (V \geq 0), \quad (15)$$

$$A_1(V) \leq \frac{\pi^{9/2}}{16 \cdot 2^{5/2}} \frac{V}{(V - 1/4)^{5/2}} \quad (V > 1/4). \quad (16)$$

In particular $A_1(V) < 4$ for all $V > 0$, and $A_1(V) < 4V^{-3/2}$ for $V \geq 1$.

Lemma 3.1. *Every integer-valued distribution with finite variance V and largest atom M satisfies $M \geq (12V + 1)^{-1/2}$. If a Bernoulli sum has integer mean m , this bounds $\mathbb{P}(X = m)$ from below.*

Proof. Adding an independent uniform variable on $[-1/2, 1/2]$ gives a density bounded by M , with variance $V + 1/12$. Among mass-one densities bounded by M , the density equal to M on the interval of length $1/M$ centered at the mean minimizes the second moment. Indeed, subtract that density and integrate against $(t - \mu)^2 - 1/(4M^2)$: both the inside and outside products are nonnegative, and the constant term integrates to zero. Its variance is $1/(12M^2)$. Darroch's integer-mean mode theorem gives the last assertion. $\square$

3.2 Sharp constants for variance at least twenty

Lemma 3.2. *If $V \geq 20$, then $C_2(V) < (3/5)V^{-3/2}$ and $A_1(V) < (1/3)V^{-3/2}$.*

Proof. For $|t| \leq 1$, $\sin(t/2) \geq t/2 - t^3/48$ and hence $2\sin^2(t/2) \geq 11t^2/24$. For $1 \leq |t| \leq \pi$ it is at least $529/1152$. The Gaussian second moment is below $(12/25)V^{-3/2}$, using $\sqrt{2\pi} > 12/5$; squaring verifies the remaining rational comparison. The other portion is at most $\pi^2 e^{-529V/1152}$. After multiplication by $V^{3/2}$ this decreases for $V \geq 20$ and is below $900/8000 = 9/80$ at twenty. Thus $12/25 + 9/80 < 3/5$.

Put $W = V - 1/4 \geq 79V/80$. The imaginary-part bound makes A_1 at most $V/6$ times a normalized fourth-moment integral. Split at 1 and $\pi/2$. The first portion, enlarged to the full Gaussian line, is below $(8/5)W^{-5/2}$. On $[1, \pi/2]$, $\sin t \geq \sin 1 > 5/6$, so

$$2\sin^2(t/2) \geq 529/1152 + (5/6)(t - 1).$$

Since $t^4 < 7$ and $\pi > 3$, this portion is below $(14/(5W))e^{-529W/1152}$. The last portion is below $100e^{-W}$. It follows that

$$A_1(V) < \frac{4}{15}VW^{-5/2} + \frac{7}{15}\frac{V}{W}e^{-529W/1152} + \frac{50}{3}Ve^{-W}.$$

After multiplication by $V^{3/2}$ these are bounded by

$$\frac{4}{15}\frac{26}{25} + \frac{7}{15}\frac{80}{79}\frac{90}{8000} + \frac{50}{3}\frac{1800}{10^8} < \frac{1}{3}.$$

Here $(80/79)^3 < 26/25$, the normalized exponential terms decrease, $529(79/4)/1152 > 9$, and $79/4 > 19$. The positive sums $\sum_{k=0}^{40} 9^k/k! > 8000$ and $\sum_{k=0}^{60} 19^k/k! > 10^8$ are exact rational witnesses. $\square$

3.3 The primitive arrays and endpoint error

For $n = d - s$, retained parameters satisfy $f_x(q/n) \geq f_x(q/d)$. The means increase in the dimension: shifting q to $q + 1$ embeds the old array beneath the new one. Thus the padded parameter distance is

$$D = \mu_n - \mu_d + 2 \sum_{q=n}^{d-1} f_x(q/d) \leq 2sp. \quad (17)$$

Put $e_j = j\nu - \mu_j$. The increasing Riemann bounds give $0 \leq e_j \leq p$. At $\mu_d = m$ the shift needed by the n-law is

$$\Delta = m - k - \mu_n = s\nu - k + \eta, \quad \eta = e_n - e_d = \mu_d - \mu_n - s\nu.$$

The trapezoid formula writes $e_j = p/2 - \epsilon_j$ with

$$|\epsilon_j| \leq \frac{|f'_x(1) - f'_x(0)|}{8j}.$$

The errors ϵ_j have the same sign for all j , because f_x is convex for $x < 1$, concave for $x > 1$, and linear for $x = 1$. Consequently $|\eta| \leq |f'_x(1) - f'_x(0)|/(8n)$, without adding two endpoint bounds. For $x \leq 1/10$ this is at most $3x/(16n)$; for $x \leq 15$ it is at most $15/(16n)$. Concavity also gives $e_j \geq p/2$ when $x \geq 1$.

4 The positive majorant and its centered moments

Define the nonnegative series

$$\mathcal{B}(u, z) = (1 + z) \sum_{n \geq 1} u^n \sum_{h|n} \frac{\varphi(h)}{h^2} (1 + z)^{n-n/h} P_{n/h}(z).$$

Its exponential majorizes the absolute coefficients of $\mathcal{H}$. At $z = x$, write

$$v_a = (1 + x)P_a(x)R^a, \quad w_{a,h} = v_a \frac{\varphi(h)}{h^2} q^{(h-1)a}, \quad \beta = \sum_{a,h} w_{a,h}, \quad \beta_1 = \sum_{a,h} h a w_{a,h}.$$

A logarithmic component has dimension $n = ha$ and degree law

$$K = X_{a,x} + \text{Binomial}((h-1)a + 1, p). \quad (18)$$

Set $\ell = (h-1)a$, $t_a = p - e_a$, $r = p - \nu$, and $T_j = \sum w_{a,h} \ell^j$. Then $0 \leq t_a \leq p$, $r \geq 0$, and

$$\mathbb{E}(K - n\nu) = t_a + \ell r.$$

After including the first extra Bernoulli, there are a parameters of total mean $a\nu + t_a$. Cauchy's inequality bounds their variance by $a\nu(1-\nu) + t_a(1-2\nu) - t_a^2/a$. Therefore

$$\text{Var } K \leq n\nu(1-\nu) + p \max(1-2\nu, 0) + \ell \{p(1-p) - \nu(1-\nu)\}. \quad (19)$$

Let $M_1 = \sum w \mathbb{E}(K - n\nu)$ and $M_2 = \sum w \mathbb{E}(K - n\nu)^2$. If $t_a \leq t_0$ and $r \leq r_0$, then

$$M_1 \leq t_0\beta + r_0T_1, \quad (20)$$

$$M_2 \leq \nu(1-\nu)\beta_1 + p \max(1-2\nu, 0)\beta + \max(0, p(1-p) - \nu(1-\nu))T_1 \\ + t_0^2\beta + 2t_0r_0T_1 + r_0^2T_2. \quad (21)$$

Write $c_{s,k} = R^s x^k [u^s z^k] \exp \mathcal{B}(u, z) \geq 0$ and $h_s = \sum_k c_{s,k}$. The exponential formula gives

$$\sum_s h_s = e^\beta, \quad \sum_s s h_s = e^\beta \beta_1, \quad (22)$$

$$\sum_{s,k} c_{s,k} (k - s\nu)^2 = e^\beta (M_2 + M_1^2), \quad (23)$$

$$e^{-\beta} \sum_{s,k} c_{s,k} |k - s\nu| \leq \min \left(\sqrt{M_2 + M_1^2}, 2\nu\beta_1 + M_1 \right). \quad (24)$$

These are infinite-series statements. First restrict to finite sets of components, for which the independent compound-Poisson calculation is exact. All coefficients and the squared weights are nonnegative; increasing component sets increases their sums. The component means are nonnegative, and the finite bounds below give finite M_1, M_2, β, β_1 . Monotone convergence proves (23). The last bound follows both from Cauchy and $|k - s\nu| \leq k + s\nu$. The divergent uncentered second *dimension* moment is never used.

Both q and v_a increase with x , since

$$\frac{d \log q}{dx} = \frac{p - \nu}{x} \geq 0, \quad \frac{d \log v_a}{dx} = \frac{p - e_a}{x} \geq 0.$$

Thus $\beta, \beta_1, T_j, b_n, h_s$ increase too, where $b_n = [u^n] \mathcal{B}(Ru, x)$. Finally,

$$\sum_n n \|J_n\|_x R^n \leq \frac{\beta_1}{1+x}, \quad E = e^{(x-1)J(R,x)} > e^{-\beta}. \quad (25)$$

5 Degree smoothing and coefficient-tail induction

Lemma 5.1. *Let $v_* = p(1-p)$ and $c(x) = \min(2/3, (1+x)^2/6)$. Each degree law of a fixed composition of components of total dimension s has variance at least $c(x)sv_*$. Every coefficient of the normalized positive majorant of dimension $s \geq 1$, even after convolution with a normalized nonnegative polynomial, is at most*

$$h_s \frac{\sqrt{\pi}}{2\sqrt{2c(x)sv_*}}.$$

Proof. The Bernoulli variance integrand is precisely

$$g_x(t) = f_x(t)(1 - f_x(t)) = \frac{xt(2-t)}{[2 + (x-1)t]^2}.$$

For $x \geq 1$ it is at least $v_*t(2-t)$; for $x \leq 1$ it is at least $(1+x)^2v_*t(2-t)/4$. Include the first extra Bernoulli as the endpoint $t = 1$, and use

$$\sum_{q=1}^a (q/a)(2 - q/a) = \frac{(a+1)(4a-1)}{6a} \geq \frac{2a}{3}.$$

The remaining $(h-1)a$ Bernoulli variables each have variance v_* , and $c(x) \leq 1$. Variances add in fixed compositions. Fourier inversion and $\sin(|t|/2) \geq |t|/\pi$ bound the largest atom of a Bernoulli sum of variance V by $\sqrt{\pi}/(2\sqrt{2V})$. A positive mixture has the same bound, and convolution is a weighted average of translates. $\square$

Lemma 5.2. *If $b_n \leq Bn^{-5/2}$ for the needed large indices and $h_j \leq Cj^{-5/2}$ is proved for the needed smaller indices, then*

$$h_s \leq s^{-5/2} \left[\frac{6\beta_1}{s} C + B e^\beta \left(1 + \frac{9\beta_1}{s} \right) \right]. \quad (26)$$

Proof. Use $sh_s = \sum_{n=1}^s n b_n h_{s-n}$. For $n \leq s/2$, the ratio $(1 - n/s)^{-5/2}$ is at most $2^{5/2} < 6$. For $n > s/2$, put $j = s-n$ and use $(1-j/s)^{-3/2} \leq 1+9j/s$ on $[0, 1/2]$. Sum $\sum h_j = e^\beta$ and $\sum j h_j = e^\beta \beta_1$. In particular bases $64 \leq j \leq 127$ suffice for induction from $s = 128$; no bound for $j < 64$ is needed in the first part. $\square$

If $v_a \leq Aa^{-5/2}$ and $q \leq q_0 < 1$, the proper divisor contributions give

$$n^{5/2} b_n \leq A + 3An^{3/2} q_0^{n/2}. \quad (27)$$

Indeed $\varphi(h)/h^2 \leq 1/h = a/n$, $a \leq n/2$ for $h \geq 2$, and $\sum_a a^{-3/2} < 3$. The right side decreases in the ranges used below.

6 A common marked-identity decomposition

Fix $d \geq 1001$, an integer $1 \leq m \leq d-2$, and its saddle $x \leq 15$. Write $\kappa = [z^m]P_d > 0$ and $S = \lfloor d/8 \rfloor$. For $n = d-s$, the primitive multiplier is

$$A_s = \frac{n}{d} \lambda^s \frac{P_n(x)}{P_d(x)} = (d/n)^{3/2} e^{\delta_n - \delta_d}. \quad (28)$$

For $s \leq S$, (10) gives

$$A_s < 5/4, \quad |A_s - 1| < \{(9/4)s + 3\}/d \quad (s \geq 1), \quad A_0 = 1. \quad (29)$$

The derivative of $(1-t)^{-3/2}$ is below $9/4$ for $t \leq 1/8$. Also $|\delta_n - \delta_d| < 1/n$ and $e^{1/n} \leq (1 - 1/n)^{-1}$; their multiplier contributes less than $3/d$. For $n \geq d/2$, the coarser bound is $A_s < 4$.

For each signed coefficient of $\mathcal{H}_s$, first compare the profiles at zero by (12), then shift the n-profile by $\Delta = m - k - \mu_n$ using (14). Divide by the integer-mean d-mode using Lemma 3.1. Thus the probability error is bounded by

$$\sqrt{12V_d + 1} \left\{ \frac{3}{2} DC_2(W) + A_1(V_n) |\Delta| + \frac{1}{2} C_2(V_n) \Delta^2 \right\}. \quad (30)$$

The intermediate n-profile at zero need not itself be a probability.

To compare (5) with E , use the small- s primitive terms, the middle primitive terms $S < s \leq d/2$, and the full J_n terms $s > d/2$. Handle $J_n - P_n$ for every $n \geq d/2$ separately. Finally bound the infinite tail of E after $s = S$ by $\sum_{s>S} h_s$. These pieces exhaust the marked identity and its comparison series; integer cutoffs use $s \leq \lfloor d/2 \rfloor$ where appropriate.

7 Small saddles

Proposition 7.1. *If $d \geq 1001$, $\mu_d(x) = m \in \{1, \dots, d-2\}$, and $0 < x \leq 1/10$, then $[z^m]\mathcal{R}_d > 0$.*

7.1 Scalar bounds and their infinite tails

At $x = 1/10$, $q = (11/20)^{20/9} < 53/200 < 4/15$, certified by ninth powers. For all smaller x , (10) and $\pi > 55/18$ imply $v_a < (3/10)a^{-5/2}$ for $a \geq 2$, while $v_1 = q < 3/10$. The exact small-saddle script certifies

$$\beta < 2/5, \quad \beta_1 < 4/5, \quad T_1 < 3/25, \quad T_2 < 1/5. \quad (31)$$

For completeness its finite mass sums use $a \leq 64, h \leq 16$. For $t = q^a$, the h tails in the mass and dimension moment are respectively $t^{16}/(17(1-t))$ and $at^{16}/(1-t)$. For $a > 64$ their total tails are bounded by $1/2560$ times $1 + q^{65}/(2(1-q^{65}))$, and $(3/40)/(1-q^{65})$. The direct periodic-moment envelopes are

$$T_1 \leq \frac{q^2}{1-q} + \frac{9}{80} \frac{q^2}{(1-q)(1-q^2)}, \quad T_2 \leq \frac{q^2}{(1-q)^2} + \frac{9}{40} \frac{q^2}{(1-q)(1-q^2)^2}.$$

The first terms handle $a = 1$; the others use $a^{-3/2} < 3/8$ and $a^{-1/2} < 3/4$ for $a \geq 2$.

Equation (27) with $A = 3/10, q_0 = 4/15$ is below $2/5$ at $n = 16$ and decreases thereafter. The exact bases $n < 16$ verify this same b_n bound. Exact $h_s^2 s^5 < 1$ for $1 \leq s \leq 23$ and (26) at $s = 24$, with $B = 2/5, C = 1, e^\beta < 3/2$, give

$$h_s < s^{-5/2} \quad (s \geq 1), \quad E > 2/3, \quad e^\beta/E < 9/4.$$

Integral bounds yield $(9/25)x \leq \nu < (2/5)x$. For the lower bound, ν/x decreases in x and at $1/10$ it equals $-10/9 + (200/81)\log(20/11)$; the positive nine-term log series with $t = 9/31$ proves the claim. For the upper bound its limit at zero is $2\log 2 - 1 < 2/5$. The integer mean forces $y = dx > 5/2$, and $V_d < (2/5)y$.

7.2 Centered losses for small components

Equations (20)–(21) give $M_1 \leq (13/25)x$ and $M_2 + M_1^2 < (24/25)x$. One explicit loose bound is $M_2 \leq (21/25)x + (21/25)x^2$: its variance's linear coefficient is at most $(2/5)(4/5) + 2/5 + (16/25)(3/25) = 0.7968$; the component squared-mean coefficient is at most $2/5 + 2(16/25)(3/25) + (16/25)^2(1/5) = 0.63552$. Together with $M_1^2 \leq (169/625)x^2$, these are below the stated bound at $x \leq 1/10$. Equation (24) also bounds the absolute first moment by $e^\beta(29/25)x$.

For $s \leq S$, $n \geq 7d/8$, so $|\eta| < x/(4d)$. Thus the weighted absolute and squared shift moments are respectively below $e^\beta(117/100)x$ and $e^\beta(97/100)x$. The weighted D bound is $2pe^\beta\beta_1 < e^\beta(8/5)x$. The prefactor error from (29), divided by E , is below $27/(4d)$.

The elementary bound $V_j \geq (j\nu - p)/(1+x)$ gives, for both needed arrays,

$$V_n \geq \frac{63}{220}y - \frac{1}{11}, \quad W \geq \frac{63}{220}y - \frac{15}{44} > 0.$$

The bounds of Section 3 imply

$$y\sqrt{(24/5)y+1}C_2(W) < 50, \quad (32)$$

$$y\sqrt{(24/5)y+1}C_2(V_n) < 50, \quad (33)$$

$$y\sqrt{(24/5)y+1}A_1(V_n) < 100. \quad (34)$$

For C_2 , split at $y = 7/2$. Below it use $C_2 < 10/3$ and the square-root bound $17/4$; above it use $W \geq 3y/16$, square root below $(23/10)\sqrt{y}$, Gaussian constant below $8/5$, and $(16/3)^{3/2} < 25/2$. For A_1 , when $y < 5$ the global bound $A_1 < 4$ gives a normalized bound below 100. When $y \geq 5$, $V_n \geq y/4$ and the bound $4V_n^{-3/2}$ gives a bound below 72, hence below 100. Using (30), the probability error divided by E is below

$$\frac{1}{d}\frac{5}{4}\frac{9}{4}\left\{\frac{12}{5}50 + \frac{117}{100}100 + \frac{97}{200}50\right\} < \frac{735}{d}.$$

Including the prefactor error gives $742/d$.

7.3 Middle terms, large terms, and periodic terms

Since $S \geq d/9$, the omitted E tail divided by E is below $27d^{-3/2}$. For $S < s \leq d/2$, $V_n \geq (9/55)y - 1/11$ and the primitive largest-atom/d-mode ratio is below 5, by squaring the Gaussian bound and using $y > 5/2, \pi < 4$. With $A_s < 4$, the middle contribution is below $540d^{-3/2}$.

For $s > d/2$, Lemma 5.1 has $c \geq 1/6$ and $v_* > (4/5)x$. Its atom bound is below $\sqrt{15/(4sx)}$. Also (10) gives $P_d(x)R^d > (5/19)d^{-5/2}$, and the inverse d-mode is below $(23/10)\sqrt{y}$. The full J_n contribution is consequently bounded by

$$\frac{35}{2}d^2 \sum_{n < d/2} n \|J_n\|_x R^n (d-n)^{-3}.$$

Use (25), $\beta_1 < 4/5$, $E > 2/3$, and $(d-n)^{-3} < 8d^{-3}$ to obtain $169/d$ after division by E .

All periodic terms with $n \geq d/2$, using (11), the same product and mode bounds, and $e^\beta/E < 9/4$, total less than $35d^3e^{-d/20} < 1/1000$. The function decreases for $d > 60$. At 1001, $e > 8/3$ and $35000 \cdot 1001^3(3/8)^{50} < 1$ are an exact witness. The overall relative error is therefore

$$\frac{912}{d} + \frac{567}{d^{3/2}} + \frac{1}{1000} \leq \frac{912}{1001} + \frac{567}{1001 \cdot 31} + \frac{1}{1000} < 1.$$

Thus $|[z^m]\mathcal{R}_d/\kappa - E| < E$, proving Proposition 7.1.

8 Larger saddles and interval certificates

Proposition 8.1. *If $d \geq 1001$, $\mu_d(x) = m$ is an integer, and $1/10 \leq x \leq 15$, then $[z^m]\mathcal{R}_d > 0$.*

8.1 Variance and uniform local comparison

The variance density is

$$v(x) = \int_0^1 \frac{xt(2-t)}{[2+(x-1)t]^2} dt.$$

For $x \neq 1$, direct integration gives

$$\nu(x) = \frac{x}{x-1} - \frac{2x \log((1+x)/2)}{(x-1)^2}, \quad (35)$$

$$v(x) = \frac{2x(x+1) \log((1+x)/2)}{(x-1)^3} - \frac{x(3x+1)}{(x+1)(x-1)^2}, \quad (36)$$

with $\nu(1) = 1/4, v(1) = 1/6$. Both ν/x and v/x decrease in x , while ν increases. The integrand g_x is nonnegative, unimodal and at most $1/4$. Its total variation is at most $1/2 - g_x(1)$, with $g_x(0) = 0$. Comparing the right Riemann sum and then subtracting the endpoint proves

$$jv - 1/2 \leq V_j \leq jv + 1/2.$$

Every certified interval has $v \geq 3/100$, so $dv \geq 30.03$. For $s \leq S$, this implies

$$W \geq (17/20)dv, \quad V_n \geq (6/7)dv, \quad \sqrt{12V_d + 1} < (7/2)\sqrt{dv}.$$

In particular $W, V_n > 20$. Lemma 3.2 yields

$$\sqrt{12V_d + 1} C_2(W), \sqrt{12V_d + 1} C_2(V_n) < \frac{27}{10dv}, \quad \sqrt{12V_d + 1} A_1(V_n) < \frac{3}{2dv}.$$

The numerical comparisons follow by squaring positive rational quantities. Section 3.3 gives $|\eta| < 1/500$, since $15/(16n) \leq 15/(16 \cdot 7 \cdot 1001/8) < 1/500$.

8.2 Whole-interval envelopes

Fix a closed interval $[a, b] \subset [1/10, 15]$ from Table 1. The endpoint helper gives rational lower and upper bounds on ν and v , and upper bounds β, B_1, T_1, T_2 at b . Set

$$v_{\text{low}} = a v(b)_{\text{low}}/b, \quad v_{\text{high}} = \min(1/4, b v(a)_{\text{high}}/a).$$

Let $N_{\text{hi}}, N_{\text{lo}}$ be the maximum and minimum of $t(1-t)$ over the enclosed ν range, and $P_{\text{hi}}, P_{\text{lo}}$ the analogous bounds over the exact endpoint p range. These extrema include a possible crossing of $1/2$. Put $p_a = a/(1+a), p_b = b/(1+b)$ and

$$r_0 = \min \left(p_b - \nu(a)_{\text{low}}, \frac{p_b(p_a - \nu(a)_{\text{low}})}{p_a} \right), \quad t_0 = \begin{cases} p_b/2, & a \geq 1, \\ p_b, & a < 1. \end{cases}$$

The second r bound follows because $(p - \nu)/p = \int_0^1 2(1-t)/[2 + (x-1)t] dt$ decreases in x . The improved t_0 follows from concavity when $a \geq 1$. Define

$$M = t_0\beta + r_0T_1, \quad (37)$$

$$Q = N_{\text{hi}}B_1 + p_b \max(1 - 2\nu(a)_{\text{low}}, 0)\beta + \max(0, P_{\text{hi}} - N_{\text{lo}})T_1 + t_0^2\beta + 2t_0r_0T_1 + r_0^2T_2 + M^2, \quad (38)$$

$$U = \min(\sqrt{Q}, 2\nu(b)_{\text{high}}B_1 + M), \quad \epsilon = 1/500. \quad (39)$$

Equations (23)–(24) bound the required shifted moments by $e^\beta(U + \epsilon)$ and $e^\beta(Q + 2\epsilon U + \epsilon^2)$.

For a lower bound L on $\log E$, use $L = -\beta$ when $a < 1$. For $a \geq 1$, set $\rho = (x-1)/(x+1)$ and note

$$\log E = \rho \sum_{j,h} v_j \frac{\varphi(h)}{h^2} (\rho q)^{(h-1)j}.$$

All factors are nonnegative and increasing in x . A positive finite sum at a is therefore a valid lower bound throughout the interval. Take certified $R_{\text{exp}} \geq e^{\beta-L}$ and $I_{\text{exp}} \geq e^{-L}$. The small-component probability and prefactor errors, divided by E , are bounded by K_{small}/d and K_{pref}/d , where

$$K_{\text{small}} = \frac{5}{4} \frac{R_{\text{exp}}}{v_{\text{low}}} \left\{ \frac{81}{10} p_b B_1 + \frac{3}{2} (U + \epsilon) + \frac{27}{20} (Q + 2\epsilon U + \epsilon^2) \right\}, \quad (40)$$

$$K_{\text{pref}} = R_{\text{exp}} \left(\frac{9}{4} B_1 + 3\beta \right). \quad (41)$$

In the first term, $(3/2) \cdot (2p_b B_1) \cdot (27/10) = (81/10)p_b B_1$; thus the full weighted parameter loss is included.

8.3 Tail bases and the remaining marked terms

The exact tail script uses the following caps at four endpoints. By monotonicity each cap applies to every smaller positive x . Here A bounds $v_a a^{5/2}$, q_0 bounds q , B bounds $b_n n^{5/2}$ for $n \geq 64$, and e_0 bounds e^β .

x	A	q_0	B	β	β_1	e_0	C
1	41/100	3/8	1/2	7/12	6/5	9/5	3/2
3	4/7	1/2	3/5	81/100	7/4	9/4	2
7	81/100	16/25	5/6	9/8	8/3	25/8	4
15	8/7	3/4	4/3	3/2	4	9/2	10

The script checks $A^2 > (1+x)5/62$ using $\pi > 31/10$, and $q_0 < A$ for $a = 1$. It checks (27) at 64 and its decreasing continuation, the exact h bases 64..127, and $6\beta_1 C/128 + B e_0(1 + 9\beta_1/128) < C$. Lemma 5.2 gives $h_s < C s^{-5/2}$ for every $s \geq 64$.

For $S < s \leq d/2$, $V_n \geq dv/2 - 1/2$; the squared Gaussian atom ratio to the d-mode is below four squared, since $dv \geq 30$. Using $A_s < 4$ and $S \geq d/9$, the middle contribution and omitted E tail together are below $306 C I_{\text{exp}} d^{-3/2}$.

For $s > d/2$, use Lemma 5.1 and (25). Set

$$c_{\text{low}} = \min(2/3, (1+a)^2/6), \quad U_P = \sqrt{4(22/7)(1+b)} \frac{1001}{1000}.$$

Equation (10) and $e^{1/d} \leq 1001/1000$ give the inverse primitive scale bound $U_P d^{5/2}$. Since $\sqrt{\pi}/(2\sqrt{2}) < 16/25$ and $s^{-3} < 8/d^3$, the full large- s contribution, divided by E , is below K_{large}/d , where

$$K_{\text{large}} = U_P \frac{7}{2} \frac{16}{25} 8 \sqrt{\frac{v_{\text{high}}}{c_{\text{low}} P_{\text{lo}}}} \frac{C B_1}{1+a} I_{\text{exp}}.$$

The square root uses the corrected smoothing variance factor.

The remaining periodic terms with $n \geq d/2$ are bounded using $P_d(x) R^d > d^{-5/2}/32$, inverse mode below $2\sqrt{d}$, $e^\beta < 9/2$, and $E^{-1} < 9/2$. Equation (11) gives

$$1296 d^3 e^{-d/20} < 1/1000 \quad (d \geq 1001).$$

This decreases for $d > 60$; at 1001 its exact endpoint certificate is $1296000 \cdot 1001^3 (3/8)^{50} < 1$, using $e > 8/3$.

The large-saddle verifier certifies, on each whole closed interval,

$$\frac{K_{\text{small}} + K_{\text{pref}} + K_{\text{large}}}{1001} + \frac{306 C I_{\text{exp}}}{1001 \cdot 31} + \frac{1}{1000} < 1. \quad (42)$$

Every term bounds all $d \geq 1001$, because the algebraic errors decrease in d and the periodic term has the monotonicity just proved. Thus this is a uniform dimension proof, not finite dimension sampling. Together with the split of Section 6, it proves $|[z^m]\mathcal{R}_d/\kappa - E| < E$ and Proposition 8.1.

Table 1: The 33 closed intervals and upward-rounded total relative error bounds from (42).

Left endpoint	Right endpoint	Certified upper envelope
1/10	1/5	0.415204
1/5	3/10	0.354745
3/10	2/5	0.337833
2/5	1/2	0.333091
1/2	3/5	0.333805
3/5	7/10	0.337634
7/10	4/5	0.343535
4/5	9/10	0.349782
9/10	1	0.356797
1	5/4	0.273062
5/4	3/2	0.269584
3/2	7/4	0.268158
7/4	2	0.268035
2	9/4	0.268862
9/4	5/2	0.270358
5/2	11/4	0.272354
11/4	3	0.274745
3	7/2	0.453021
7/2	4	0.450379
4	9/2	0.450842
9/2	5	0.453294
5	11/2	0.457061
11/2	6	0.461875
6	13/2	0.467545
13/2	7	0.473930
7	8	0.911391
8	9	0.904558
9	10	0.903392
10	11	0.906514
11	12	0.912996
12	13	0.922190
13	14	0.933627
14	15	0.946958

9 Exact endpoint arithmetic and reproducibility

This section specifies the analytic tails and enclosure rules used by the certificates. None of the scalar tests replaces a preceding analytic proof. For a rational endpoint x ,

$$q = \left(\frac{1+x}{2}\right)^{-2/(x-1)} \quad (x \neq 1), \quad q(1) = e^{-1}.$$

Exact rational-power bisection encloses q in 48 steps. The logarithm of a positive rational a is enclosed using $t = (a-1)/(a+1)$, 128 terms, and remainder $2|t|^{257}/(257(1-t^2))$. For exponentials use 97 positive terms (indices 0..96) and the upper remainder $a^{97}/(97!(1-a/98))$ for $0 \leq a < 20$; negative arguments use reciprocals. Signed linear combinations in (36) use the correct endpoint order. Upward and downward rounding use the dyadic lattice of denominator 2^{72} . An upward square-root bound is

$$\frac{\lfloor \sqrt{[a2^{100}]} \rfloor + 1}{2^{50}}.$$

No floating-point value controls a logical decision.

For the large endpoint sums take $a \leq 64$ and $h \leq H_a = \max(2, \lfloor 128/a \rfloor)$. With $t = q^a$, the four omitted h tails for β, β_1, T_1, T_2 , before multiplication by v_a , are

$$\frac{t^{H_a}}{(H_a+1)(1-t)}, \quad \frac{at^{H_a}}{1-t}, \quad \frac{at^{H_a}}{1-t}, \quad a^2 t^{H_a} \left(\frac{H_a}{1-t} + \frac{t}{(1-t)^2} \right).$$

Let $A > \sqrt{(1+x)/(4\pi)}$ also bound v_1 and put $t = q^{65}$. The omitted a tails are at most

$$\frac{A}{768} \left(1 + \frac{t}{2(1-t)} \right), \quad \frac{A}{4(1-t)}, \quad \frac{A}{512} \frac{t}{(1-q)(1-t)}, \quad \frac{A}{8} \frac{t}{(1-q)(1-t)^2}.$$

The first two follow from integral sums after 64; the others use $a^{-3/2} \leq 1/512$, $a^{-1/2} \leq 1/8$ and geometric sums. For the positive lower log-E sum take $a \leq 32$ and $h \leq \max(1, \lfloor 32/a \rfloor)$, discarding only nonnegative terms.

The three scalar programs use only Python's standard library:

```
verify_small_x_majorant.py
verify_majorant_tail_constants.py
verify_large_x_intervals.py
```

They are preserved byte for byte from the reviewed candidate. The second imports the third's enclosure helpers. All checks raise explicit exceptions; they do not use Python `assert`. The certificates give respectively the small-saddle scalar/base checks, four finite-base/induction residuals, and 33 whole-interval rational bounds. Their exact rational entries and SHA256 digests accompany this paper. The 6-decimal numbers in Table 1 are rounded *upwards* from already-certified rational bounds. The largest exact bound, on $[14, 15]$, is

$$\frac{2235941074644793697351}{2361183241434822606848} < 0.947.$$

The reproducibility entry point also runs the independent scalar residual checks, verifies file integrity, and compares the generated witnesses. Floating-point displays are diagnostics only.

The new proof was internally reviewed with AI assistance and its three exact programs reproduced independently as part of the accompanying audit. This record is not Lean verification or external human peer review. The source audit maps every necessary candidate argument to the final sections; historical incomplete labels and a corrected scratch variance expression occur only in the separately identified historical review record.

10 Completion of the proof

Proof of Theorem 1.1. For $d \geq 1001$, Proposition 2.1 covers $r \leq \lfloor d/4 \rfloor$. For the remaining needed coordinates put $m = d - 1 - j$, so

$$1 \leq m \leq d - 2 - \lfloor d/4 \rfloor < 3d/4.$$

The last strict inequality holds for every residue class of d modulo four. The elementary estimate $\mu_d(15) \geq d\nu(15) - 15/16 > 3d/4$ for $d > 1000$ follows from $\log 8 < 209/100$ and $\nu(15) = 15/14 - (15/98)\log 8 > 751/1000$. Thus each required unique saddle lies below 15. Propositions 7.1 and 8.1 cover all these saddles, including their common endpoint $1/10$. Hence $a_{d,j} > 0$ for $\lfloor d/4 \rfloor + 1 \leq j \leq d - 2$. Every remaining tail (1) is a positive sum of these coordinates. The prior endpoint facts and repair criterion give real zeros. Since the Ehrhart coefficients are nonnegative and $H_d(0) = 1$, $H_d(t) > 0$ for $t \geq 0$, so all its zeros are strictly negative.

For $d \leq 1000$, apply de Castro's finite Theorem 1.3. Its base cases include $H_1 = H_2 = 1$ and $H_3 = (1 + t)^2$. This covers all dimensions. $\square$

10.1 The imported finite certificate

The finite theorem uses the author's official release `tashimir/ehrhart-132-213-polytopes`, release `v1.0.0`, published on 5 September 2026. The file is `refined_coordinates_d1_d1000.csv`, of size 996682200 bytes. The official release asset digest is

```
8dbb43cad35659199cd23b5fddc23dd50
f76d35adff5b636ac366d460069b5ce
```

A complete streamed download independently confirmed this byte length and SHA256 digest, and counted 500501 lines including the header. The accompanying provenance record distinguishes byte-integrity verification from mathematical reconstruction. We have not independently reconstructed the CSV or rerun the source's 498500 finite tail inequalities. Theorem 1.1 depends transparently on that prior finite theorem.

11 A corollary for integral dilates

Corollary 11.1. *For every pair of positive integers d, r , the polynomial $h^*(rP_d; t)$ is constant or has only strictly negative real zeros.*

Proof. For $d \geq 3$, P_d has dimension $d - 1$ in its affine lattice, so the Ehrhart series has denominator $(1 - t)^d$. The Veronese construction gives

$$h^*(rP_d; t) = U_{r,0}^d H_d(t), \quad U_{r,k}^d h = \left((1 + t + \cdots + t^{r-1})^d h(t) \right)^{\langle r,k \rangle}.$$

Here $f(t) = \sum_{k=0}^{r-1} t^k f^{\langle r,k \rangle}(t^r)$ defines the sections. By Theorem 1.1, H_d has nonnegative coefficients and only negative real zeros. Zhang's Corollary 3.4 [4, p. 7] applies for every positive integer r , with no degree threshold on r , and makes the sequence $(U_{r,r-1}^d H_d, \dots, U_{r,0}^d H_d)$ interlacing. In particular the zeroth member is real-rooted. Its nonnegative coefficients and constant term one exclude nonnegative roots. For $d = 1$ the numerator is 1; for $d = 2$ the primitive lattice segment gives $1 + (r - 1)t$ directly. This is an application of the established Veronese theorem to this family, not a new preservation theorem. $\square$

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```
python3 reproduce.py --negative-controls
```

The computational checks do not replace the analytic proof.

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